

Complex Analysis

5.5 Harmonic function: Laplace Equation

$$f(z) = u + iv, \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}, \quad \frac{\partial^2 u}{\partial x^2} = \frac{\partial^2 v}{\partial x \partial y}, \quad \frac{\partial^2 u}{\partial y^2} = -\frac{\partial^2 v}{\partial y \partial x}$$

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \Rightarrow \frac{\partial^2 u}{\partial x^2} = -\frac{\partial^2 u}{\partial y^2}$$

$$\Rightarrow \nabla^2 u = 0 \quad \text{similarly,} \quad \Rightarrow \nabla^2 v = 0$$

So if u, v follow the Laplace equation it means they form a harmonic conjugate.

Example: $u = x^2 - y^2 - y$ & $v = 2xy + x$, is they form harmonic conjugate or not?

Solution: $\frac{\partial u}{\partial x} = 2x, \frac{\partial^2 u}{\partial x^2} = 2$ & $\frac{\partial u}{\partial y} = -2y - 1, \frac{\partial^2 u}{\partial y^2} = -2$

$$\frac{\partial v}{\partial x} = 2y + 1, \frac{\partial^2 v}{\partial x^2} = 0 \quad \& \quad \frac{\partial v}{\partial y} = 2x, \frac{\partial^2 v}{\partial y^2} = 0$$

Since $u_{xx} + u_{yy} = 0$ & $v_{xx} + v_{yy} = 0$ so they form a harmonic conjugate.

Milne Thomson's Method:

To write $f(z)$ when u and v is known, Milne Thomson's method will be used.

$$f(z) = u + iv$$

(a) If u is given:

$$\phi_1 = \frac{\partial u}{\partial x}, \quad \phi_2 = \frac{\partial u}{\partial y}, \quad f(z) = \int \phi_1(z, 0) dz - i \int \phi_2(z, 0) dz$$

(b) If v is given

$$\psi_1 = \frac{\partial v}{\partial y}, \quad \psi_2 = \frac{\partial v}{\partial x}, \quad f(z) = \int \psi_1(z, 0) dz + i \int \psi_2(z, 0) dz$$

Example: $u = x^2 - y^2 - y, \Rightarrow f(z) = ?$

$$\phi_1 = \frac{\partial u}{\partial x} = 2x \Rightarrow \phi_1(z, 0) = 2z, \quad \phi_2 = \frac{\partial u}{\partial y} = -2y - 1 \Rightarrow \phi_2(z, 0) = -1$$

$$f(z) = \int 2z dz + i \int 1 dz = \frac{2z^2}{2} + iz = z^2 + iz$$

$$f(z) = z(z + i) = (x + iy)^2 + i(x + iy) = x^2 - y^2 + 2ixy + ix - y = (x^2 - y^2 - y) + i(2xy + x)$$

Example: $v = 2xy + x, f(z) = ?$

$$\psi_1 = \frac{\partial v}{\partial y} = 2x \Rightarrow \psi_1(z, 0) = 2z, \quad \psi_2 = \frac{\partial v}{\partial x} = 2y + 1, \quad \psi_2(z, 0) = 1$$

$$f(z) = \int 2z dz + i \int 1 dz = \frac{2z^2}{2} + iz = z^2 + iz$$

$$f(z) = z^2 + iz = (x^2 - y^2 - y) + i(2xy + x)$$

Example: Given an analytic function $f(z) = \phi(x, y) + i\psi(x, y)$. What will be the value of ψ if

$$\phi = x^2 + 4x - y^2 + 2y?$$

Solution: $\phi = u = x^2 + 4x - y^2 + 2y, \quad \psi = v = ?$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 2x + 4 \Rightarrow v = 2xy + 4y + g(x), \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -2y + 2 \Rightarrow v = \int (2y - 2) \partial x$$

$$v = 2xy - 2x + h(y)$$

$$v = 2xy + 4y - 2x$$

Example: If $z^2 = x^2 - y^2 + 2ixy$ then $\lim_{z \rightarrow 0} \frac{\operatorname{Re} z^2 + \operatorname{Im} z^2}{z^2} = ?$

Solution: $\lim_{z \rightarrow 0} \frac{\operatorname{Re} z^2 + \operatorname{Im} z^2}{z^2} = \lim_{z \rightarrow 0} \frac{x^2 - y^2 + 2xy}{x^2 - y^2 + 2ixy} = \lim_{x+iy \rightarrow 0} \frac{x^2 - y^2 + 2xy}{x^2 - y^2 + i2xy}$

$$\lim_{x \rightarrow 0} \frac{-y^2}{-y^2} = 1, \quad \lim_{y \rightarrow 0} \frac{x^2}{x^2} = 1, \quad \lim_{x=y} \frac{x^2 - x^2 + 2x^2}{x^2 - x^2 + i2x^2} = \frac{1}{i} = -i$$

$\therefore$ Limits are different, therefore limit does not exist, and the function is non-analytic.

Example: Which of the following functions given below, cannot be a real part of a complex analytic function.

- (a) $x^2 y$ (b) $x^2 - y^2$ (c) $x^3 - 3xy^2$ (d) $3x^2 y - y - y^3$

Solution: (a) $u = x^2 y \Rightarrow \frac{\partial u}{\partial x} = 2xy, \quad \frac{\partial u}{\partial y} = x^2, \quad \frac{\partial^2 u}{\partial x^2} = 2y, \quad \frac{\partial^2 u}{\partial y^2} = 0$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2y \neq 0, \text{ So } u = x^2 y \text{ cannot be a real part of a complex analytic function.}$$

(b) $u = x^2 - y^2 \Rightarrow \frac{\partial u}{\partial x} = 2x, \quad \frac{\partial^2 u}{\partial x^2} = 2, \quad \frac{\partial u}{\partial y} = -2y, \quad \frac{\partial^2 u}{\partial y^2} = -2$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ So } u = x^2 - y^2 \text{ can be a real part of a complex analytic function.}$$

(c) $u = x^3 - 3xy^2, \quad \frac{\partial u}{\partial x} = 3x^2 - 3y^2, \quad \frac{\partial^2 u}{\partial x^2} = 6x, \quad \frac{\partial u}{\partial y} = -6xy, \quad \frac{\partial^2 u}{\partial y^2} = -6x$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ So } u = x^3 - 3xy^2 \text{ can be a real part of a complex analytic function.}$$

(d) $u = 3x^2 y - y - y^3 \Rightarrow \frac{\partial u}{\partial x} = 6xy, \quad \frac{\partial^2 u}{\partial x^2} = 6y, \quad \frac{\partial u}{\partial y} = 3x^2 - 1 - 3y^2, \quad \frac{\partial^2 u}{\partial y^2} = -6y$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \text{ So } u = 3x^2 y - y - y^3 \text{ can be a real part of a complex analytic function.}$$

Example: Check whether the following functions are analytic or not :

(a) $f(z) = \frac{\bar{z}}{z}$

Solution: $f(z) = \frac{\bar{z}}{z} = \frac{x-iy}{x+iy} = \frac{(x-iy)(x-iy)}{(x+iy)(x-iy)} = \frac{(x-iy)^2}{x^2+y^2}$

$$f(z) = \frac{x^2 - y^2 - 2ixy}{x^2 + y^2} \Rightarrow u = \frac{x^2 - y^2}{x^2 + y^2}, \quad v = \frac{-2xy}{x^2 + y^2}$$

$$\frac{\partial u}{\partial x} = \frac{(x^2 + y^2)2x - (x^2 - y^2)2x}{(x^2 + y^2)^2} = \frac{2x(x^2 + y^2 - x^2 + y^2)}{(x^2 + y^2)^2} = \frac{4xy^2}{(x^2 + y^2)^2}$$

$$\frac{\partial v}{\partial y} = \frac{-2x(x^2 + y^2) + 2xy(2y)}{(x^2 + y^2)^2} = \frac{2x(2y^2 - x^2 - y^2)}{(x^2 + y^2)^2} = \frac{2x(y^2 - x^2)}{(x^2 + y^2)^2}$$

$\frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}$, So this is a non-analytic function.

Or

$$f(z) = \frac{x-iy}{x+iy}, \quad \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z} = \lim_{\Delta z \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y}$$

$$\lim_{\Delta x \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = -1, \quad \lim_{\Delta y \rightarrow 0} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = 1, \quad \lim_{\Delta x = \Delta y} \frac{\Delta x - i\Delta y}{\Delta x + i\Delta y} = \frac{1-i}{1+i}$$

All are different so the function is not differentiable.

(b) $f(z) = \frac{|z|^2}{z}$

Solution: $f(z) = \frac{|z|^2}{z} = \frac{z\bar{z}}{z}$

$$f(z) = \frac{|z|^2}{z} = \frac{z\bar{z}}{z} = \frac{x^2 + y^2}{x+iy} = \frac{x^2 + y^2(x-iy)}{(x+iy)(x-iy)} = \frac{(x^2 + y^2)(x-iy)}{(x^2 + y^2)}$$

$$f(z) = x - iy \Rightarrow u = x, v = -y, \quad \frac{\partial u}{\partial x} \neq \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = \frac{-\partial v}{\partial x}$$

So Non-analytic function.

Example: Check whether the following functions satisfy the $C-R$ equation or not.

(a) $f(z) = z^2$

Solution: $f(z) = z^2 = (x+iy)^2 = x^2 - y^2 + 2ixy, \quad u = x^2 - y^2, v = 2xy$

$$\frac{\partial u}{\partial x} = 2x, \frac{\partial v}{\partial y} = 2x, \quad \frac{\partial u}{\partial y} = -2y, -\frac{\partial v}{\partial x} = -2y, \text{ C-R equation satisfied.}$$

(b) $f(z) = z^3$

Solution: $f(z) = (x + iy)^3 = x^3 - iy^3 + 3x^2iy - 3xy^2$, $u = (x^3 - 3xy^2)$, $v = (3x^2y - y^3)$

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} = 3x^2 - 3y^2 \quad \& \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} = -6xy$$

C-R equation satisfied.

(c) $f(z) = e^{2x}(\cos y + i \sin y)$

Solution: $u = e^{2x} \cos y$, $v = e^{2x} \sin y$, $\frac{\partial u}{\partial x} = 2e^{2x} \cos y$, $\frac{\partial v}{\partial y} = e^{2x} \cos y$

$$\frac{\partial u}{\partial y} = -e^{2x} \sin y, \quad -\frac{\partial v}{\partial x} = -2e^{2x} \sin y, \quad \text{So the C-R equation is not satisfied.}$$

Example: Let $u(x, y) = e^{ax} \cos by$ be the real part of the function

$$f(z) = u + iv, \quad z = x + iy, \quad a, b \in R, \quad a \neq 0$$

The function $f(z)$ is complex analytic everywhere in the complex plane if

- a) $b = 0$
- b) $b = \pm a$
- c) $b = \pm 2\pi a$
- d) $b = a \pm 2\pi$

Solution: $u(x, y) = e^{ax} \cos by$,

From Laplace equation, $u_{xx} + u_{yy} = 0$, $u_{xx} = -u_{yy}$

$$\frac{\partial u}{\partial x} = ae^{ax} \cos by, \quad \frac{\partial^2 u}{\partial x^2} = a^2 e^{ax} \cos by, \quad \frac{\partial u}{\partial y} = -be^{ax} \sin by, \quad \frac{\partial^2 u}{\partial y^2} = -b^2 e^{ax} \cos by$$

$$a^2 e^{ax} \cos by = +b^2 e^{ax} \cos by \Rightarrow a^2 = b^2 \Rightarrow \pm a = b$$

Infinite series: